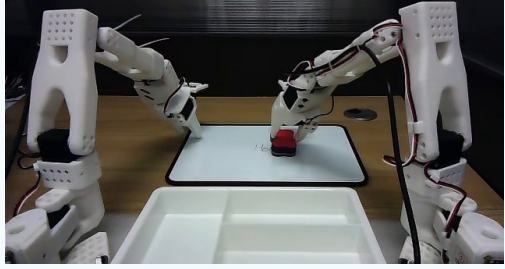


Inference: structured bimanual VLA

Synchronized inputs

RGB $\times 3$



Global Left wrist Right wrist

Language

“Wipe the board.”

Joint state

$q \in \mathbb{R}^{12}$

Vision
encoder

Visual tokens

Z_v

Language
encoder

Language tokens

Z_l

State
encoder

State tokens

Z_s

Shared task context

$Z \in \mathbb{R}^{N \times d}$

Dual-arm VLA

Joint policy π_θ

Joint action chunk

32 steps $\times$ 12 dimensions

Left: 5 joints
+ gripper

Right: 5 joints
+ gripper

$a_{1,1}$	$\cdots$	$a_{1,6}$	$a_{1,7}$	$\cdots$	$a_{1,12}$
$a_{2,1}$	$\cdots$	$a_{2,6}$	$a_{2,7}$	$\cdots$	$a_{2,12}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$a_{32,1}$	$\cdots$	$a_{32,6}$	$a_{32,7}$	$\cdots$	$a_{32,12}$

Absolute joint targets

Gradient update

Training-only relation supervision

Predicted actions

$\hat{a} \in \mathbb{R}^{32 \times 12}$

Demonstrated actions

$a^* \in \mathbb{R}^{32 \times 12}$

Training context

q_t + geometry
Contact / Role labels (fixed)

Shared context, mapping and statistics

Relation map + normalization

Pose $\cdot$ Distance

$\hat{\rho}$

Relation map + normalization

Pose $\cdot$ Distance

ρ^*

Weighted
comparison

L_{rel}

Total loss

$$L = \underbrace{L_{\text{BC}}}_{\text{Imitation}} + \underbrace{\lambda_{\text{rel}} L_{\text{rel}}}_{\text{Relation}} + \underbrace{\lambda_{\text{safe}} L_{\text{safe}}}_{\text{Safety prior}}$$